

Dynamic Difficulty Adjustment by performance and player profile in platform game

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Abstract. The Dynamic Difficulty Adjustment (DDA) of games can play an important role in increasing the player engagement and fun. This work investigates the different mechanisms of a DDA system for a platform game to adequately adapt its difficulty level and keep the player in a state of flow. The proposed adjustment varies the size of the platform and the height of the jump, comparing different approaches from the game systems. Tests were made with sample groups, in which participants answered questionnaires and had their data collected for research purposes. The results indicated that the difficulty of platform games can be estimated by the components of the levels, including correlation between the difficulty and player performance data. In addition, player profiles were predicted from raw game session data and used with machine learning methods to define difficulty progression. Finally, the DDA models were able to adjust the game difficulty to the players, decreasing the dispersion between the performance data and keeping the player in a state of flow, especially when using a feed forward neural network to progress difficulty based on the player's profile.

Keywords: dynamic difficulty adjustment · platform game · player profile · machine learning · game telemetry · flow

1 Introduction

The requirements and rules to make good games need to adapt to the new scenarios, creating fun games for each consumer profile. New mechanisms are required to cover a diversity of players and allow games to target users with different skill levels and preferences.

Boredom or frustration can be caused by non-adaptive games by not assessing the match between the degree of challenge and the player profile [3]. In contrast to the pre-established difficulty levels, the Dynamic Difficulty Adjustment (DDA) is an emerging technique that seeks to adapt the difficulty of a game during its execution according to the skills of each player [1].

There is a field of study in the DDA area to define robust metrics and rules for adaptation in relation to the concept of flow and the pattern of game systems

[17][2]. Furthermore, few studies apply machine learning outside the context of NPC adaptation or associate adaptation with player characteristics other than their ability, such as specific characteristics of player profiles.

For the present work, the objective is to increase the robustness of the metrics for measuring difficulty and component analysis, generalizing them to be used in platform games. For the construction of the proposed model, game balancing will be analyzed in relation to generic components of platform games and the corresponding difficulties. Also, the efficiency of computational methods for estimating the difficulty of individual components and game levels will be investigated.

Thus, the estimated difficulty will be compared with that experienced by the players, allowing a better assessment of the DDA in all analyzed variations. With this, the models will be compared, finding out which ones have comparatively superior results and if all are significantly better than the game without applying adjustments. For this, the state of flow and the adequacy of the difficulty will be evaluated, according to questionnaire responses and the dispersion of game data captured during test sessions.

Considering the objectives presented, a 2D platform game will be adapted according to the most common components of this genre. In addition, the game will be adjusted in order to progress according to the difficulties experienced by each player, trying to keep it in a suitable difficulty range.

The game will have a DDA that analyzes the player's performance by heuristic functions or by machine learning, relating the progression of difficulty with the player's profile. Furthermore, this work will present the correlation between raw game data with the difficulty experienced and the player's profile, in addition to the use of characteristics related to difficulty from the player profiles to improve the DDA method.

In a second instance, tests will evaluate the DDA applied to configure the size of platform components and jump height. Finally, tests will investigate the use of machine learning and the association of player profiles with the difficulty experienced, verifying the metrics for measuring the flow state and the adequacy of the difficulty in studies with the application of DDA. In this scenario, a comparison will be made between the DDA with heuristic functions and with an artificial neural network (ANN).

2 Related Work

Difficulty in automatically generated 2D platform games has been investigated by Mourato et al [12]. A method of measuring difficulty based on the probability of success in completing a level was proposed for this context, with the difficulty of individual components made by a jump success probability model.

From there, commercial games had their difficulty mapped and the results of difficulty measured and tested with participants were similar. Also, regarding content generation [11], the transformation of simple paths into challenges with complex structures was elaborated, changing components in strategic locations.

In turn, the dynamic balancing of difficulty in a puzzle game was the subject of study by Hawkins et al [5]. They sought to meet different skill levels and maintain satisfaction according to the characteristics of the player's profile, depending on the affinity for challenges and risks. A modeling technique known as particle filtering was used, being applied to model various skill levels and to consider the risk profile of the player.

In the context of machine learning, neural networks can be used to improve adaptability proposals in games. Or et al [13] proposed a DDA method that optimized the user experience by considering other players and game rules. It was based on a deep neural network that minimized the loss of user experience function. Furthermore, experiments in a puzzle game outperformed manual heuristics made by game designers.

On the other hand, the work presented by Rosa et al [14] had the development of a hybrid DDA, including performance and affective data, for a platform game. This work investigated if the DDA mechanism appropriately adapted its difficulty level and kept the player in a state of flow. The three approaches (performance, affective and hybrid) were compared to verify the efficiency of each model, with the affective state of the player estimated through the electrodermal activity (EDA) of the skin.

The results indicated that the three DDA models were able to adjust the gameplay difficulty to the players when varying platform size. Specifically about the performance-based DDA system, the variation of platform size and of jump height were tested independently and together, verifying the adequacy of difficulty in the adaptations. However, the improvement of the player's state of flow was not verified.

3 Methodology

The present work deals with the development of a DDA system for platform games. For this, it was performed the improvement of difficulty estimation methods, the definition of a technique to recognize player profiles and the application of DDA through categorized player profiles.

Thus, based on the studies of [4] and [2] a cyclic diagram of the DDA model was made (Figure 1). There are four interconnected systems that refer to the raw data of the Player, the Monitoring of these metrics, and the corresponding Analysis and Control Systems of the components that will be adjusted.

This work updates the measurement of the difficulty based on platform components [12], and revisits the concepts and results for a system-based DDA [14], noting that all systems must be independently tested. For this context, metrics were determined to measure the flow state [6][9] and the player profile [8], based on the literature review and analysis of the participants' responses.

Several adaptations were made to the open-source game *The Explorer: 2D*, developed by *Unity Technologies*. Changes in the game were made to create levels, adjust game components to the proposal, implement communication with external tools, distribute for multiple devices, and enable game adaptation:

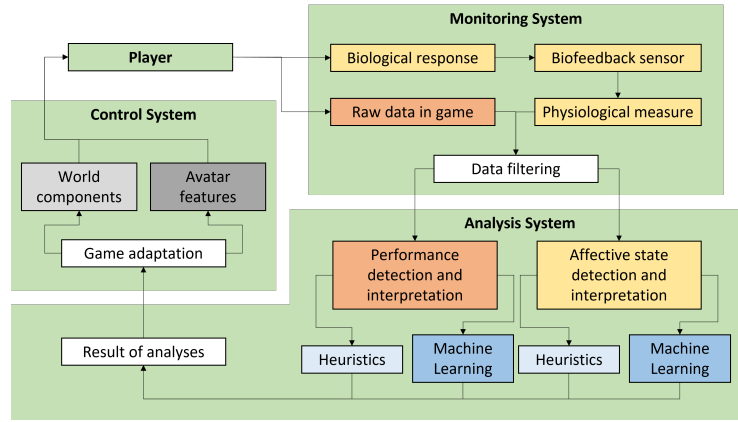


Fig. 1. Cyclic diagram of the system-oriented DDA.

- Control System: adaptation of platform sizes on the horizontal and vertical axes (Figure 2); of the avatar’s jump height (Figure 3); and of the combinations of these two adaptations.
- Analysis System: analysis based on heuristics, considering the difficulty experienced by the player; and based on the machine learning method by feed forward ANN, considering the difficulty experienced and the player’s profile.



Fig. 2. Illustration of platform size adaptation, showing the original level and the adaptation that decreases the difficulty level.

In addition, the efficiency of computational methods that estimate the game difficulty was evaluated. For that, the platform difficulty was computed from the probability of success of a jump [14][12]. The jump corresponds to a parabola and the probability of success comes from the horizontal and vertical error of the parabola from an origin point in relation to a given end point. Besides, based on equations of kinematic physics, new jump configurations were included: jump impulse speed, variation of angle or direction in the jump, and platforms with an end before the point of the jump parabola destination, as shown in Figure 3.

Further, it was sought to cover the difficulty for game components other than platforms. These difficulties were calculated by the inverse of the sum of the com-

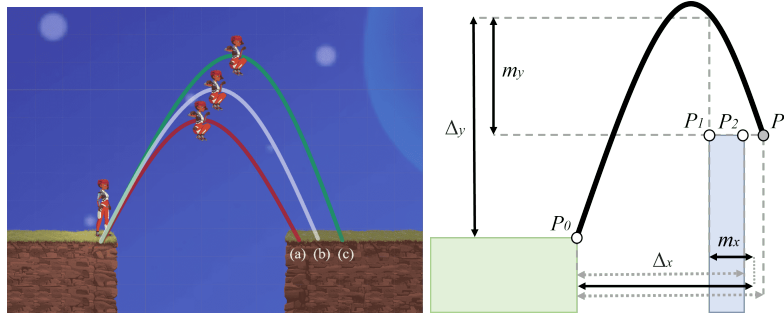


Fig. 3. Illustration of the difference in difficulty for jump height, being (a) minimum jump height, (b) medium and (c) maximum. Followed by an updated graphical representation of the jump's margin of error.

ponent's characteristic attributes, such as enemy health and attack range, the space occupied by a trap, and the distance to be covered by a mobile object. Having measured each individual component difficulty, the level difficulty is calculated by multiplying the average component difficulty and an inverse proportion of the number of components, prioritizing the global mean while still considering the level size. In this way, the method is able to calculate the total difficulty of a level for any platform game, associating generic components of multiple platform sub-genres (platforms, obstacles, movement aids, collectibles and triggers), as described by Smith [15].

Considering the robustness of the DDA method, rules were defined for the Control System, such as the equal prioritization of changes on the x and y axes, and the change of origin according to the nearest point. Also, adaptations must have spatial limits to maintain design standards, avoid conflicts in the map, and prevent the existence of unreachable components.

The focus of the study was on tailoring the difficulty to each player and keeping it in a state of flow when investigating DDA models in platform games. It is observed that smaller platforms or shorter distances between platforms, and smaller jump heights represent greater difficulty. With that, the game was modified based on the data related to the prediction of the difficulty experienced by the player, and the general progression of the difficulty was fixed for a heuristics model and made from the categorization of the player between player profiles of the Hexad model [10] for a ANN model.

While data was captured and analyzed in real time, the adaptation was applied only before the start of the level or when the player died to avoid the DDA from being too noticeable to the player. If the player experience was considered harder than the desired for the level by the Analysis System, the Control System adapted the level to be easier, and vice versa. Also, the individual components were adapted by adding the difference between its estimated difficulty and the experienced difficulty to prevent outliers, such as ten easy components and one very difficult.

In both models, the experienced difficulty was determined based on death rate, execution of successful jumps by the player, and time played in a level. In the heuristics model, this difficulty varies according to the sum of the inverse of the number of levels (for the difficulty progression) raised by the inverse of the three input variables (to express the difficulty based on the player performance).

The ANN model used a supervised learning method for classification. Thus, a feed forward ANN was encoded with three attributes in the input layer and one attribute in the output layer (experienced difficulty or player profile). ANN training was performed with the backpropagation algorithm and the sigmoid activation function, considering the data scale always between 0 and 1 and running 1000 times with the learning rate updated according to the sum of squared errors.

Moreover, the Hexad model associates systems with user personalities and is based on studies on human motivation and types of players, having a significant association with the Big Five global personality factors model [7]. In this context, there are six types of profiles [16]: Achiever (motivated by mastery); Socialiser (by relatedness); Philanthropist (by purpose); Free Spirit (by autonomy); Player (by reward); and Disruptor (by change).

The player profile was limited to the Free Spirit or Achiever profiles of the Hexad model. Considering that players of the Achiever profile seek mastery, they prefer a greater challenge in the game compared to Free Spirit players. Thus, the player profile was determined considering proximity to Achiever over Free Spirit from input data of the amount of collectibles gathered, the execution of ranged attacks compared to melee attacks, and the time walking towards the level exit, without exploration. This value was used to determine how much the difficulty should progress when starting each level in the ANN model.

4 Experiments

Three experiments were conducted, with remote execution (whether on a computer or mobile phone) and participation of players with different skill levels and familiarity with games. The first test intended to verify the improvement made in the difficulty estimation method that involves more platform configurations and other components of the genre. In addition, the objective was to investigate the efficiency of computational methods to recognize the player profiles of the Hexad model and the difficulty experienced in this work, statistically correlating the extracted game data with the actions of the players.

Therefore, the game contained 8 levels, testing different components and difficulty situations for platform games. The levels were planned to contain generic platform game components in order to ascertain their difficulty. Fifty-one volunteers aged between 15 and 55 years participated in the experiment. Additionally, data from this test were used to train the neural network.

The second test aimed to apply more robust metrics to the performance-based DDA, verifying whether the system keeps the player in a state of flow and has a better result in adapting the difficulty to the user. The comparison tests

of DDA Control System models needed to stimulate different reactions, testing metrics and perception instruments.

For this, the game presented 10 levels with extremes of difficulty, as the questionnaires related to the state of flow did not have significant differences between the DDA models tested in [14]. That is, by presenting levels alternating between easy and difficult, it is expected to avoid co-variance between the difficulty progression of the level design and the adaptations performed by the Control System.

Four adaptation models in the Control System were compared: (1) without the application of DDA, (2) with platform size adaptation, (3) with jump height adaptation and (4) with combined platform size and jump height adaptation. 112 volunteers participated in this experiment, 28 for each model tested, with ages ranging from 15 to 55 years. In this context, the following hypothesis was verified for the proposed models:

Theorem 1. *A DDA system combining platform and jump height adaptation provides adequate difficulty and keeps the player in a state of flow.*

In the third test, the objective was to use machine learning to improve the DDA Analysis System by recognizing the player profile and estimating the difficulty. Thus, it was verified the ability of a DDA system to integrate machine learning and use characteristics of the player's profile to keep the player in a state of flow and reduce the dispersion between the performance of different players.

The game presented 8 levels with extremes of difficulty and three models of the Analysis System were compared: (1) without the application of DDA, (2) with analysis based on heuristics and (3) with analysis based on machine learning. 69 volunteers participated, with 23 testing each model and ages ranging from 18 to 45. It is noted that only one model in the Control System was applied, being the combined DDA of platform size and jump height adaptation. Thus, the following hypothesis was defined:

Theorem 2. *A DDA system with machine learning, that defines the adaptations by the difficulty experienced and by the player's profile, provides adequate difficulty and keeps the player in a state of flow.*

All tests had questionnaires, applied at the beginning and end of the game session to, respectively, capture sociodemographic data and verify the hypotheses of each experiment. The first level of each test was a tutorial, with due attention to present different components and without the application of DDA. Furthermore, each experiment had different volunteers and the test conditions were specified beforehand to ensure that participants had the same information and chances to complete the game.

5 Discussion

5.1 Investigation of difficulty estimation and player data

In the first experiment, the player was asked to describe the difficulty experienced in the level numerically in values from 0 to 10 at the end of each level, as shown

in Figure 4. The estimated difficulty and that obtained by the average informed by the players have less difference between their values in the new method, with the estimated values always within the first and third quartiles reported by the participants. Thus, it shows that they maintain a significant proportion, with results superior to those presented by the previous method of [14].

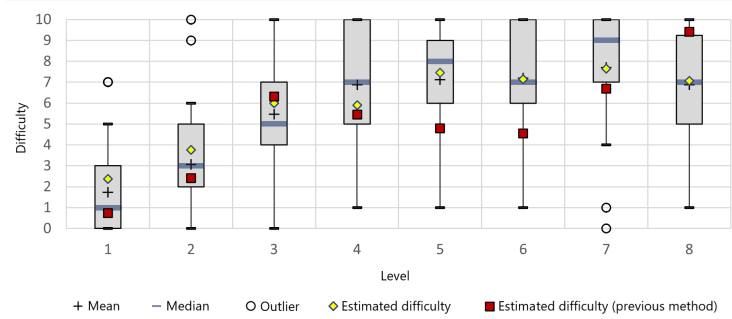


Fig. 4. Box-plot with the average difficulties informed in the questionnaire and the values estimated by algorithms in the first experiment.

It is also observed that the percentage of participants who complete a level decreases in cases of high difficulty or with the presence of puzzles, and the DDA aims to reduce the number of dropouts. Moreover, other metrics were measured at the end of each level, following the standard Likert scale with values between 1 and 7. Among the data analyzed, it was found that when too easy or too difficult, boredom and frustration increased for the average player, needing to maintain a standard of difficulty according to the player's skills.

For this, a selection of resources supervised by filtering with numerical input and categorical output (Free Spirit or Achiever) was performed using ANOVA selection. All game data were tested and the most relevant statistical tests obtained to predict the profiles were: percentage of collectibles gathered, time walking in the opposite direction to the end and number of ranged attacks.

These actions are related to the exploration and mastery profile, with the player using the game time to perform more activities in the level and to carry out actions in a planned way. Therefore, it was possible to identify performance variables that are significantly related to Free Spirit or Achiever profiles, which were used in the neural network.

Additionally, a supervised selection of resources was made with data related to the difficulty experienced by the player, with numerical input and output (difficulty between 0 and 1) and using the Pearson selection. All game data were tested and the most relevant statistical tests were obtained for: percentage of successful jumps, time played and death rate. Thus, these parameters can predict the difficulty experienced by players, usable in heuristics and in the neural network of the DDA Analysis System for subsequent tests.

5.2 Comparison between DDA Control System models

In the second experiment, the adequacy of the difficulty was confirmed in relation to the decrease in the dispersion of the performance of different players. In this sense, the variation of all data collected from the game was investigated, observing that the averages improved and had less deviation when a DDA model was used, as seen in Figure 5.

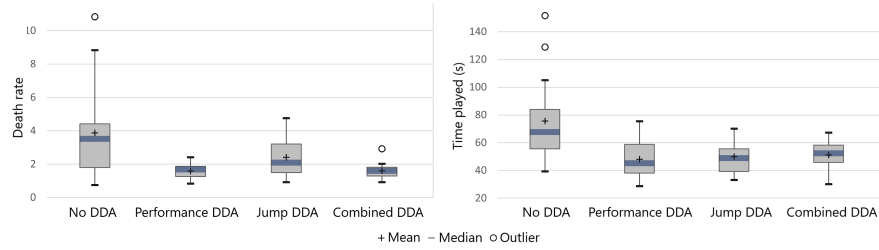


Fig. 5. Box-plot with average death rate and time played per level by participants in the second experiment, divided by the DDA Control System model.

The death rate and time played averages were always smaller and the successful jumps average higher when applying some DDA model, especially the platform DDA. While the performance improvement is generally positive for players, this is not a direct indication that the model is better than others. A game can only facilitate so that it always gets better results from the players. In turn, the decrease in dispersion between performances is a more complex factor to be achieved, as it is directly dependent on the players' skills.

To compare the means between the 4 groups, the ANOVA technique was used, with a significance level of 0.05. Thus, all values obtained significant variation between means, with the hypothesis that the difficulties are equal rejected in multiple cases, such as the proportion of successful jumps, the death rate and the average time played for each level by the participants. It shows that the combined DDA was the one that presented the lowest deviation among all the models for all the collected data, followed by the platform DDA, the jump DDA and, in the end, without DDA.

Moreover, there was a higher number of completed levels (ending without using the function to give up the level) when using combined DDA (98%) compared to platform DDA (96%), jump DDA (90%) and without DDA (87%). This indicates that the application of performance adjustments can be associated with players' resilience.

At the end of the game session, questions were asked about the players' experience, difficulty adequacy and flow state. It was noted that the means were significantly different when using the ANOVA technique with a significance level of 0.05. First, it was verified how the players described their own performance, with the best averages for the platform (5.52) and combined (5.42) DDA than

of the game without DDA (4.47). For participants with a predilection for games with easy difficulty, the difference was more significant when comparing the game without DDA (2.66) with the platform (4.25), jump (5) and combined (5) DDA.

The greatest fun happened in the game with platform DDA (5.47), followed by the jump DDA and combined DDA (5.05), and without DDA (4.64). Also, the lowest boredom and frustration were with the platform DDA (2.87; 2.47), followed by the combined DDA (3.10; 3.10), without DDA (3.35; 3.94), and the jump DDA (3.88; 4.17). It is noted that the jump DDA may not be ideal for use in future tests as it had a negative impact on users' responses, while the other models demonstrate consistency.

To compare the difficulty, it was asked whether the game suited the user's skills and the difficulty increased progressively. The results were worse for the game without DDA (3.17; 4.11) and better for the platform DDA (5.26; 5.47). When analyzing the flow state, it was asked if the player worried about failing, had control of his actions, lost track of time, had automatic actions, and was focused on the game. The means without DDA (5.41; 3.88; 2.88; 4.82; 3.88) were worse in all questions, with the best results varying for each question between models, mostly in the combined DDA (4.10; 5.79; 4.05; 5.47; 5.53).

Theorem 1 was confirmed by providing adequate difficulty and keeping players in a state of flow in a game with a model combining adaptation of platform size and jump height. In this sense, there was significance in the difference between means for the flow state, which had not occurred in the tests of [14]. That is, the adaptations were not causing significant variance in the results related to the flow state in such tests due to the stimulus provided by the levels not having a significant difference. This change was successful and made it possible to corroborate the flow state measurement format through questionnaires and game data.

5.3 Comparison between DDA Analysis System models

In the third experiment, the suitability of the difficulty for the different models of the Analysis System was verified. Thus, all data collected from the game was analyzed in order to identify if there was less dispersion for the DDA models. The data that had the greatest decrease in dispersion and improvement in the average of the participants were for the percentage of successful jumps, percentage of collectibles gathered, death rate and time played, shown in Figure 6.

To compare the means between the 3 groups, the ANOVA technique was used, with a significance level of 0.05. All values obtained significant variation between means, and the hypothesis that the difficulties are equal was rejected. Specifically on the death rate and the time played, it had a greater reduction in the average to complete each level and a smaller dispersion when the ANN model was applied compared to the other models.

One data that improved only with the neural network DDA was the percentage of collectibles gathered, which associates with the adequacy of the challenge progression by the player's profile and correlated motivation to complete additional tasks in the levels. Furthermore, there was a higher number of completed

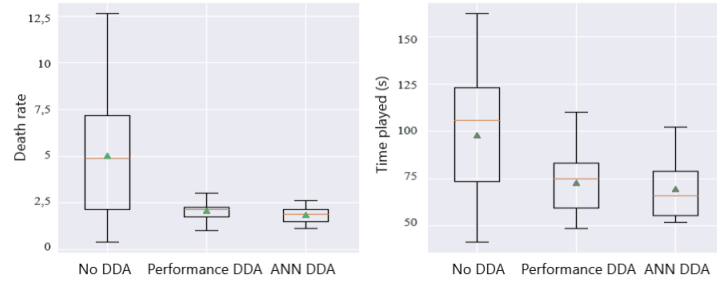


Fig. 6. Box-plot with average death rate and time played per level by participants in the third experiment, divided by the DDA Analysis System model.

levels when using the ANN model (98%) compared to the heuristics model (95%) e without DDA (87%).

Analyzing specific profiles, it was noted that the percentage of successful jumps is higher for players of the Achiever profile when no DDA is applied and that this average increases for both profiles (Achiever and Free Spirit) when a DDA model is applied. It is also observed that the heuristics DDA maintains a wide variation in the percentage of collectibles gathered for the Free Spirit profile, although there is improvement for the Achiever profile. However, this is improved and have less dispersion for both profiles by applying neural network DDA.

The same questionnaire of the second experiment was applied at the end of the game, with all the means being significantly different when using the ANOVA technique with a significance level of 0.05. In first instance, it was investigated how the players described their own performance, with the ANN model having a higher average (6.36) than the heuristics (5.65) and without DDA (4.24).

Regarding difficulty, the dispersion was lower and the results were better for the ANN model referring to adequacy and progression (6.53; 6.05) compared to the heuristics model (5.83; 5.81) and without DDA (3.12; 2.92). The questions referring to the state of flow (worry about failing, control of actions, loss of track of time, automatic actions, and focus on game) were also worse in all questions without DDA (5.23; 3.37; 2.79; 3.80; 4.01), with the best results being for the ANN model (2.81; 6.34; 6.56; 6.72; 6.61) than of the heuristics model (3.22; 5.43; 5.66; 5.87; 5.04).

Fun occurred more in the game with RNA DDA (6.53), than by the heuristics DDA (6.06) and without DDA (4.12). Moreover, the lowest boredom and frustration were with the RNA DDA (2.12; 2.24), the heuristics DDA (2.12; 2.54) and without DDA (4.00; 5.36). In the flow theory, people have different flow zones according to their ability, with no ideal path of unique difficulty in the activity. As the DDA varies according to the player's profile, it allows a more precise fit of the player in his flow zone.

The considerations presented above confirmed Theorem 2, with a better adjustment of the difficulty and maintenance of the flow state. Adequacy can be

noted by player performance statistics, including lower dispersion among players, and by higher average responses on questionnaires. Both models were successful in their proposal, but there were superior results with the neural network Analysis System adapting the progression by the player's profile.

6 Conclusion

Studies of Dynamic Difficulty Adjustment are growing in scope of themes and works carried out. However, few researches adapt as part of a game system or include characteristics of player profiles for adaptation. Therefore, this work proposed improvements to methods of difficulty estimation and player profile prediction for usage in a DDA system.

The focus was on tailoring the difficulty to each player and keeping it in a state of flow when investigating DDA models in platform games. With that, the game was modified based on the data related to the prediction of the difficulty experienced by the player and the general progression of the difficulty was made from the categorization of the player between the Free Spirit and Achiever profiles of the Hexad model.

From the analyzed results, the DDA system was capable of adapting the difficulty and keeping the player in a state of flow for a platform game, with reduced performance dispersion among players and improved responses given by participants regarding flow and difficulty adequacy and progression, with a significant difference between means. In particular, the difficulty was found to be more adequate when using models that only adapt platforms or that combine the adaptation of platforms and the jump height.

In turn, it was found that a DDA system, with analysis performed by a feed forward ANN, is capable of adjusting the difficulty and keeping the player in a state of flow for a platform game when applying characteristics of the player profile for progression of difficulty. In this context, a decrease in the dispersion of player performance and an improvement in the answers regarding the flow and difficulty were observed, with a significant difference between means and better results for the model with neural network than with heuristics analysis. It was also verified that performance can be superior with DDA, including a higher frequency of players completing levels.

Future work include adaptation with affective state captured by different physiological signals and the usage of machine learning techniques for analysis. Furthermore, other neural network models and machine learning methods can be tested and compared to predict the player profile and the difficulty experienced.

The difficulty can be also analyzed as influenced by the design of the game, with the investigation of puzzles, specific missions and content that challenge the player. Finally, adaptability can be applied in other contexts, such as players' health, where DDA can be integrated in studies that apply games to rehabilitation, memory training or increase of attention, for instance.

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